DATA ANALYST

Advanced Excel (Very Important)

You must know:

- **VLOOKUP, HLOOKUP, XLOOKUP**
- **INDEX + MATCH**
- **Pivot Table, Pivot Chart**
- **Conditional Formatting**
- **Data Cleaning**
- **Text functions (LEFT, RIGHT, MID, TRIM)**
- **IF, Nested IF, COUNTIF, SUMIF**
- **Removing duplicates**
- **Basic dashboards in Excel**

✅ 2. SQL (Most Important Skill)

You must be very strong in:

- **SELECT, WHERE, ORDER BY, GROUP BY**
- **JOINS (INNER, LEFT, RIGHT, FULL)**
- **Subqueries**
- **Aggregate functions (SUM, COUNT, AVG)**
- **HAVING**
- **CASE WHEN**
- **Date functions**
- **String functions**

✅ 3. Power BI or Tableau (Any One)

You must know:

- **Data import**
- **Data cleaning in Power Query**
- **Data modeling (relationships)**
- **DAX basics (SUM, CALCULATE, FILTER)**
- **Charts and dashboards**
- **KPI cards**
- **Slicers**

- Publishing reports

✅ 4. Python (Pandas – Basic Only)

You must know:

- pandas
- read_csv, head, info
- Filtering data
- Groupby
- Handling missing values
- Basic plots (matplotlib)

✅ 5. Statistics (Only Required Part)

You do NOT need full statistics.

Just know:

- Mean, Median, Mode
- Standard deviation
- Correlation
- Outliers
- Basic probability

✅ 6. Data Cleaning & Data Understanding

You must be able to answer:

"What steps will you take when a raw dataset is given?"

- Remove nulls
- Remove duplicates
- Correct data types
- Handle outliers

✅ 7. Real Projects (Very Important)

You must have at least 4–5 projects like:

- Sales dashboard
- HR analytics dashboard
- Student performance analysis
- E-commerce data analysis

INTERVIEW QUESTION

◆ Excel Interview Questions

1. What is a Pivot Table? Where is it used?
2. Difference between VLOOKUP and XLOOKUP
3. What is INDEX + MATCH?
4. What is Conditional Formatting?
5. How to remove duplicates?
6. What is the use of IF, COUNTIF, SUMIF?
7. How do you clean data in Excel?
8. What is data validation?
9. What is a dashboard in Excel?
10. Difference between Pivot Table and normal table

◆ SQL Interview Questions (Very Important)

1. Difference between WHERE and HAVING
2. What is GROUP BY?
3. Types of JOIN with example
4. Difference between INNER JOIN and LEFT JOIN
5. What is a Subquery?
6. What are Aggregate functions?
7. What is PRIMARY KEY and FOREIGN KEY?
8. Write query to find 2nd highest salary
9. Write query to remove duplicate rows
10. What is CASE WHEN?
11. What is ORDER BY?
12. Difference between DELETE, DROP, TRUNCATE
13. What is normalization?
14. What is indexing?

◆ Power BI / Tableau Questions

1. **What is Power BI?**
2. **What is DAX?**
3. **What is data modeling?**
4. **What is relationship between tables?**
5. **What are slicers?**
6. **What is KPI card?**
7. **What is Power Query?**
8. **Difference between Power BI and Excel**
9. **Explain your dashboard project**
10. **What steps you followed to create dashboard?**

- **Python (Pandas) Questions**

1. **What is pandas?**
2. **How to read a CSV file?**
3. **What is DataFrame?**
4. **How to handle missing values?**
5. **What is groupby?**
6. **How to filter rows?**
7. **Difference between list and DataFrame**
8. **Basic plotting in Python**

- **Statistics Questions**

1. **Mean, Median, Mode difference**
2. **What is standard deviation?**
3. **What is correlation?**
4. **What are outliers?**
5. **What is probability?**

- **Data Analytics Basic Questions (Very Common)**

1. **What is Data Analytics?**
2. **What is Data Cleaning?**
3. **What steps you follow when dataset is given?**

4. Difference between Data Analyst and Data Scientist
5. What tools you know?
6. Explain one of your projects
7. How do you handle null values?
8. Why should we hire you as a data analyst fresher?

🧠 Practical Questions They May Give

- Create a Pivot Table from data
- Write SQL query on the spot
- Explain a dashboard you made

PRACTICAL QUESTION

🧪 Practical 1 — Excel (Sales Data)

Tomake ekta data deya holo:

OrderID	Product	Region	Sales	Date
101	Laptop	East	50000	01-01-25
102	Mouse	West	2000	03-01-25
103	Laptop	East	52000	05-01-25
104	Keyboard	South	3000	07-01-25
105	Mouse	East	2500	08-01-25

Task:

1. Total sales ber koro
2. Region wise sales ber koro (Pivot Table)
3. Highest selling product ta ber koro
4. East region er total sales
5. Sales > 3000 filter koro

🧪 Practical 2 — SQL Query

Table: employees

emp_id	name	dept	salary
1	Amit	HR	30000
2	Ravi	IT	50000
3	Sita	IT	55000
4	John	HR	35000
5	Riya	IT	60000

Write SQL queries for:

1. **IT department er sob employee dekhao**
2. **Average salary of IT dept**
3. **Highest salary employee**
4. **2nd highest salary**
5. **Department wise total salary (GROUP BY)**

🧪 Practical 3 — Data Cleaning (Interview favourite)

Tomake ekta dataset deya holo jekhane:

- **Null values ache**
- **Duplicate rows ache**
- **Date column text format e**
- **Salary column e ₹ sign ache**

Question:

Ki ki step nibe data clean korar jonno?

🧪 Practical 4 — Power BI / Dashboard Thinking

Tomake bola holo:

“Create a Sales Dashboard for company”

Ki ki visual nibe?

Bolo:

- **Kon chart use korbe?**
- **Kon KPI dekhabe?**
- **Slicer ki nibe?**

🧪 Practical 5 — Python (Pandas)

CSV file ache students.csv

Columns: name, marks, city

Task:

1. CSV read koro
2. Marks > 80 filter koro
3. Average marks ber koro
4. City wise student count (groupby)

Roadmap

Goal: Excel + SQL + Power BI + Python + Projects + Resume

Study time: 3–4 hours daily

📅 Month 1 — Excel (Foundation)

Week 1

- Excel basics, tables, sorting, filtering
- IF, COUNTIF, SUMIF
- Text functions: LEFT, RIGHT, MID, TRIM

Week 2

- VLOOKUP, XLOOKUP
- INDEX + MATCH
- Remove duplicates, data cleaning
- Conditional formatting

Week 3

- Pivot Table
- Pivot Chart
- Data validation
- Dashboard basics

Week 4

- Create Sales Dashboard in Excel (Project 1)

✅ Outcome: You can pass Excel test in interview

📅 **Month 2 — SQL (Most Important)**

Use: MySQL / PostgreSQL

Week 1

- **SELECT, WHERE, ORDER BY**
- **AND, OR, NOT, LIKE**

Week 2

- **GROUP BY, HAVING**
- **Aggregate functions**

Week 3

- **JOINS (INNER, LEFT, RIGHT, FULL)**
- **Subqueries**
- **CASE WHEN**

Week 4

- **Practice 100+ SQL questions**
- **Create Employee Database Project (Project 2)**

✅ **Outcome: Strong in SQL interview**

📅 **Month 3 — Power BI**

Week 1

- **Data import**
- **Power Query (cleaning)**
- **Relationships**

Week 2

- **Charts, visuals**
- **Slicers, KPI cards**

Week 3

- **DAX basics: SUM, CALCULATE, FILTER**

Week 4

- **Create Sales Dashboard in Power BI (Project 3)**
- **Create HR Dashboard (Project 4)**

✅ **Outcome: You can explain dashboards confidently**

📅 Month 4 — Python + Projects + Resume

Week 1

- pandas basics
- read_csv, filtering, groupby
- Handling missing values

Week 2

- Student data analysis project (Project 5)

Week 3

- Revise Excel + SQL + Power BI
- Prepare interview questions

Week 4

- Make resume
- Apply daily on LinkedIn, Naukri, Indeed

✅ **Outcome: Interview ready**

🧩 Final Project List (Must Have)

1. Excel Sales Dashboard
2. SQL Employee Database queries
3. Power BI Sales Dashboard
4. Power BI HR Dashboard
5. Python Student Analysis

🎯 After 4 Months You Can Apply For:

- Data Analyst
- MIS Executive
- Reporting Analyst
- BI Analyst (fresher)